RODRIGO GOMES DE ARAÚJO

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Education

Master's — **Sep 2025 – Present**
Computer Science — *University of Porto (FCUP)*

Bachelor's — **Sep 2022 – Jul 2025**
Computer Science and Engineering — *University of Porto (FEUP)*

Experience

Murph | *Co-founder & Lead Engineer* | *Rust, gitoxide, gRPC, Ed25519, IPLD* — **Jul 2026 – Present**

- Co-founded a **3-person startup** (Top 10 of 130 at Unicorn Factory Lisboa) and wrote most of the codebase.
- Built a platform where **AI agents and humans** ship changes together, merging version control, tasks and CI.
- Encoded the merge pipeline as **Rust typestates**, so no change reaches main without verification and review.
- Designed **Ed25519-signed approvals**, tiered authority for agents and humans and a tamper-evident history.
- Built a Git layer with **compare-and-swap** ref updates, so agents and verification run in parallel worktrees.
- Implemented **crash-safe persistence** that recovers state from Git via atomic checkpoints.
- Served reads through a **byte-weighted LRU cache** of content-addressed Git objects, needing no invalidation.

Research

CaST (Cryptographically-aware Session Types) | *Master's Thesis* — **Jul 2026 – Present**

- Extending session types to track **cryptographic resource usage**, rejecting protocol misuse at compile-time.
- Formalizing the extension with a **soundness proof** and integrating it into the FreeST compiler (Haskell).
- Built a **Kademlia DHT in FreeST** and a syntax highlighting extension for the Zed editor.

Projects

PoW Blockchain for Decentralized Auctions | *Data Systems Security*, 19/20 — *Rust, Tokio, libp2p*

- Surveyed Bitcoin, Ethereum, Solana and **churn and attack research** to guide network and consensus design.
- Built a peer-to-peer overlay on a **Kademlia DHT** (as in Ethereum's discovery layer) for peer discovery.
- Designed a **Proof-of-Work blockchain** with BLAKE2b, Ed25519, Merkle trees and longest-chain consensus.
- Created an auction client that uses the chain as a public ledger for recording and verifying bids.
- Simulated bootstrap, honest and malicious nodes in Docker to study adversarial behaviour.

Distributed Protocols | *Distributed Systems*, 19/20 — *Rust, Tokio, Tonic*

- **Totally ordered multicast** with Lamport clocks, hardened against reused, backdated and far-future timestamps.
- **Token-ring mutual exclusion** with failure detection and automatic ring healing.
- **Push-pull gossip** with exponential anti-entropy for state convergence.

Portuguese Elections Data Platform | *Advanced Topics in Databases*, 18/20 — *PostgreSQL, Python, OCaml*

- Built an **ETL pipeline** normalizing 8 elections, reconciled against official totals and **D'Hondt** recomputed in SQL.
- Designed a **star-schema warehouse** with cube/rollup, window functions and drill-down to 3,000+ parishes.
- Served vote-swing analytics and PostGIS maps through an **OCaml** API to a Bonsai UI.

Achievements and Awards

Top 10 of 130 Startups, Fall 2026 Academia Cohort with Murph | Unicorn Factory Lisboa — **Oct 2026**

2nd Place, IEEE RetroJam (48-hour game in Rust) | IEEE UP Student Branch — **Oct 2025**

1st Place, National Robotics Championship; RoboCup 2016 Participant | IP Bragança — **May 2016**

Technical Skills

Languages: Rust, Haskell, OCaml, C/C++, Java, Python, SQL

Distributed Systems & Cryptography: Tokio, libp2p, Kademlia, gRPC, gitoxide, Ed25519, BLAKE2b, Merkle Trees

Data & Infrastructure: PostgreSQL, PostGIS, FastAPI, Docker, Terraform, Git, UNIX/Linux

Web: React, Dioxus, Bonsai, Dream, HTML5, CSS

Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)